

Unit One: Biology and Technology

1.1 Learning from nature

After the successful completion of this section, the student will be able to:

- Learn technologies from nature
- Recognize devices copied from biological nature
- Design devices from biological structural mechanisms

1.2 Biology and technology

After the successful completion of this section, the student will be able to:

- Describe the use of technology in biology
- Explain biology and technology
- Explain the benefits of biology to technology

1.3 Impacts of biology and technology on society and the natural

After the successful completion of this section, the student will be able to:

- Describe the impacts of biology and technology on society and nature
- Explain the relationships between biology and technology
- Determine the negative and positive impacts of biology and technology on the society and on nature

1.4 Ethical issues in biology

After the successful completion of this section, the student will be able to:

- Define ethical issues in biology
- Explain ethical issues in medicaments
- Respect ethical issues in biology

Unit Two: Animals

2.1. Characteristics of animals

After the successful completion of this section, the student will be able to:

- List the characteristics of animals
- Explain the common characteristics of animals

2.2. Invertebrates and Vertebrates

After the successful completion of this section, the student will be able to:

- List the main characteristics of invertebrates and vertebrates.
- Compare invertebrates with vertebrates' characteristics.

2.3 Reproduction in Animals

After successful completion this section, the student will be able to:

- Examine reproduction in invertebrate and vertebrate animals.
- Discuss complete and incomplete metamorphosis.
- Describe how rats, birds, frogs, and crocodiles reproduce.
- Compare and contrast reproduction in rats, birds, frogs, and crocodiles.

2.4 The economic importance of animals (insects)

After the successful completion of this section, the student will be able to:

- Describe the economic importance of insects in agriculture and food production
- Explain the economic importance of insects to industry
- Explain the economic importance of insects in health and medicine

2.5 Animal Behavior

After the successful completion of this section, the student will be able to:

- Define animal behavior
- Describe the differences between innate and learned animal behaviors.
- Identify patterns of behavior in animals

2.6 Homeostasis in animals

After the successful completion of this section, the student will be able to:

- Define homeostasis, homeotherms, poikilotherms, and thermoregulation.
- Describe thermoregulation in homeothermic animals with examples.
- Explain thermoregulation in poikilothermic animals with examples.
- Explain osmoregulation and sugar regulation.
- Discuss the mechanisms of controlling homeostasis.
- Discuss the physiological methods of thermoregulation with examples.
- Discuss the behavioral methods of thermoregulation with examples.

2.7 Renowned zoologists in Ethiopia

After successful completion of this section, the student will be able to

- Mention renowned zoologists in Ethiopia
- Describe the work of these scientists